Solution 1 Skew Howe Duality

Problem 1 — Let V, W be complex vector spaces. Show that

$$\wedge^n(V \otimes W) \cong \bigoplus_{|\lambda|=n} S^\lambda V \otimes S^{\lambda^\dagger} W$$

as $GL(V) \times GL(W)$-modules.

Idea (Hint) — Repeat the proof of the usual Howe duality (Subsection 29.2), using Corollary 30.10.

$\wedge^n(V \otimes W)$ is naturally identified with the subspace of $(V \otimes W)^{\otimes n}$ which is antisymmetric under the action of S_n. Equivalently, this is the subspace of $V^{\otimes n} \otimes W^{\otimes n} \otimes \mathbb{C}_{sgn}$ which is *invariant* under S_n. Thus, by Schur-Weyl duality, we can expand as $GL(V) \times GL(W) \times S_n$-modules:

$$\begin{aligned}
V^{\otimes n} \otimes W^{\otimes n} \otimes \mathbb{C}_{sgn} &\cong \left(\bigoplus_{|\lambda|=n} S^\lambda V \otimes \pi_\lambda \right) \otimes \left(\bigoplus_{|\mu|=n} S^\mu W \otimes \pi_\mu \right) \otimes \mathbb{C}_{sgn} \\
&= \bigoplus_{|\lambda|=|\mu|=n} S^\lambda V \otimes S^\mu W \otimes \pi_\lambda \otimes \pi_\mu \otimes \mathbb{C}_{sgn} \\
&= \bigoplus_{|\lambda|=|\mu|=n} S^\lambda V \otimes S^\mu W \otimes \pi_\lambda \otimes \pi_{\mu^\dagger} \\
&= \bigoplus_{|\lambda|=|\mu|=n} S^\lambda V \otimes S^\mu W \otimes \mathbb{C}^{\delta_{\lambda,\mu^\dagger}} \\
&= \bigoplus_{|\lambda|=n} S^\lambda V \otimes S^{\lambda^\dagger} W
\end{aligned}$$

as desired.

Solution 2 Characters and Dimensions of Irreducible Representations

Problem 2 — Compute characters and dimensions of irreducible representations $L_{a+b,b,0}$ of $SL_3(\mathbb{C})$, where $a, b \geq 0$. Compute the weight multiplicities and draw the weights on the hexagonal lattice for $a + b \leq 3$, indicating the multiplicities. What are the special features of the case $b = 0$?

Idea — The best way to do this exercise is to compute the characters recursively, using that

$$V \otimes L_{a+b,b,0} = L_{a+b+1,b,0} \oplus L_{a+b,b+1,0} \oplus L_{a+b-1,b-1,0}$$

(if $a = 0$, the second summand drops out and if $b = 0$ then the third one drops out), by the "addable boxes" rule. This allows one to express the characters for $b + 1$ in terms of the characters for b and $b - 1$. And we know the characters of $L_{a,0,0}$ - they are the complete symmetric functions h_a.

Instead of applying the hint, we're just going to use Weyl Character formula.

$$s_\lambda(x_1, x_2, x_3) = \frac{\det(x_i^{\lambda_j+N-j})}{\prod_{i<j}(x_i - x_j)}$$

We can write out the full expression. The numerator is

$$\det \begin{pmatrix} x_1^{\lambda_1+2} & x_2^{\lambda_1+2} & x_3^{\lambda_1+2} \\ x_1^{\lambda_2+1} & x_2^{\lambda_2+1} & x_3^{\lambda_2+1} \\ x_1^{\lambda_3} & x_2^{\lambda_3} & x_3^{\lambda_3} \end{pmatrix}$$

Plugging in $\lambda = (a + b, b, 0)$, we get

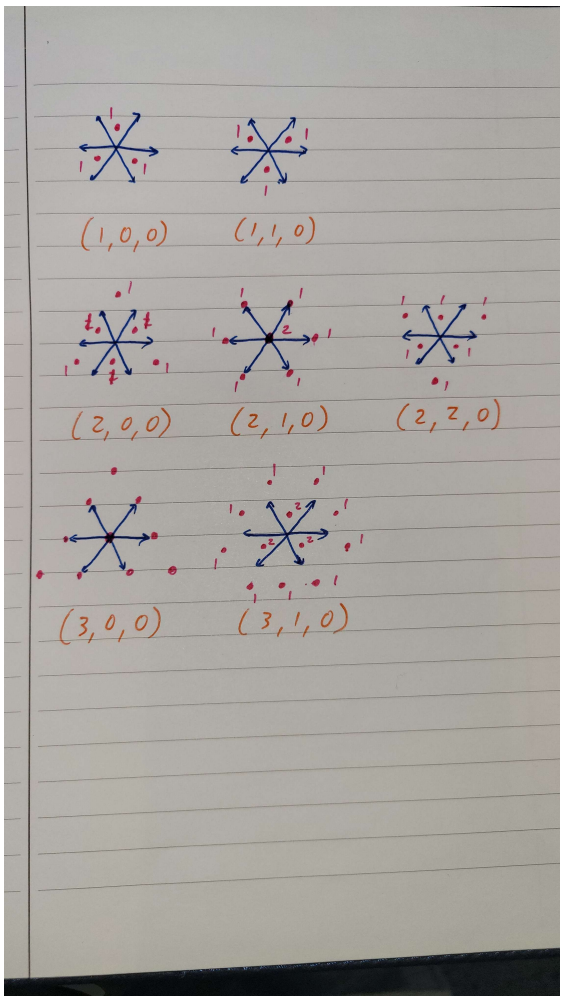
$$s_{a+b,b,0}(x_1, x_2, x_3) = \frac{\det \begin{pmatrix} x_1^{a+b+2} & x_2^{a+b+2} & x_3^{a+b+2} \\ x_1^{b+1} & x_2^{b+1} & x_3^{b+1} \\ 1 & 1 & 1 \end{pmatrix}}{(x_1 - x_2)(x_1 - x_3)(x_2 - x_3)}$$

We could simplify this easily, but this is just more tedious calculations. Moving right along, the Weyl Dimension Formula is

$$\dim L_\lambda = \frac{\prod_{i<j}(\lambda_i - \lambda_j + j - i)}{\prod_{i<j}(j - i)}$$

In our case, we have

$$\dim L_{a+b,b,0} = \frac{(1 + a)(1 + b)(2 + a + b)}{1 \cdot 2 \cdot 1} = \frac{(1 + a)(1 + b)(2 + a + b)}{2}$$



I did not draw the 1-dimension representation (a single point at the origin with dimension 1) or $L_{3,2,0}$ (which is $L_{3,1,0}$ but flipped) and $L_{3,3,0}$ (which is $L_{3,0,0}$ but flipped).

Solution 3 Decomposition of Tensor Products

Problem 3 — Compute the decomposition of

$$\wedge^m V \otimes S^k V, \quad \wedge^m V \otimes \wedge^k V, \quad S^2(\wedge^m V), \quad \wedge^2(\wedge^m V)$$

into irreducible representations of $GL(V)$.

We know that $\wedge^m V \otimes L_\lambda = \bigoplus_{\mu \in \lambda + m} L_\mu$, where we sum over partitions obtained by adding m addable boxes to different rows. We can use this to compute the decomposition of $\wedge^m V \otimes S^k V$ and $\wedge^m V \otimes \wedge^k V$:

$$\wedge^m V \otimes S^k V = \bigoplus_{\mu \in (k,0,\dots,0)+m} L_\mu = L_{(k+1)(1^{m-1})} \oplus L_{(k)(1^m)}$$

and

$$\wedge^m V \otimes \wedge^k V = \bigoplus_{\mu \in (1^k)+m} L_\mu = \sum_{i=0}^{\min(m,k)} L_{(2^i)(1^{m+k-2i})}$$

The other two are a bit more difficult. Let me explicitly construct the highest weight vectors inside of $\wedge^m V \otimes \wedge^m V$. (We assume that the dimension of V is sufficiently large; else some things simply vanish).

$$\sum_{T \subset \{\ell+1,\dots,2m-\ell\}, |T| = m-\ell} \operatorname{sgn}(T) e_1 \wedge \cdots \wedge e_\ell \wedge e_{T_1} \wedge \cdots \wedge e_{T_{m-\ell}} \otimes e_1 \wedge \cdots \wedge e_\ell \wedge e_{\bar{T}_1} \wedge \cdots \wedge e_{\bar{T}_{m-\ell}}$$

(here, T is a subset of $\{\ell+1, \dots, 2m-\ell\}$ of size $m-\ell$, and $\bar{T}$ is its complement; these are sorted, and $\operatorname{sgn}(T)$ is the parity of the sum of the elements in T).

This guy is obviously the highest weight of the representation $L_{(2^\ell)(1^{2m-2\ell})}$. Indeed, consider acting on this object to raise $\alpha+1$ to α. If $\alpha \le \ell$, this immediately vanishes. Else, if α and $\alpha+1$ are both in T, or both in $\bar{T}$, the term vanishes. Else, suppose $T = (\dots, \alpha, \dots)$ and $\bar{T} = (\dots, \alpha+1, \dots)$. This term in the sum pairs with $T = (\dots, \alpha+1, \dots)$ and $\bar{T} = (\dots, \alpha, \dots)$, where we swap only α and $\alpha+1$. This flips the sign, and both elements are raised to the same value, thus the whole sum vanishes.

These are all of them, because

$$(\wedge^m V)^{\otimes 2} = \bigoplus_{\ell \le m} L_{(2^\ell)(1^{2m-2\ell})}$$

In addition, it is symmetric if and only if $m-\ell$ is even. We conclude that

$$S^2(\wedge^m V) = \bigoplus_{\ell \le m, m-\ell \text{ even}} L_{(2^\ell)(1^{2m-2\ell})}$$

and

$$\wedge^2(\wedge^m V) = \bigoplus_{\ell \le m, m-\ell \text{ odd}} L_{(2^\ell)(1^{2m-2\ell})}$$